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RESEARCH ARTICLE

An Efficient and Robust Approach Using Inductive Transfer-Based Ensemble Deep Neural Networks for Kidney Stone Detection

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ABSTRACT Chronic kidney disorder is a global health problem involving the repercussions of impaired kidney function and kidney failure. A kidney stone is a kidney scenario that impairs kidney function. Because this disease is usually asymptomatic, early and quick detection of kidney problems is essential to avoid significant consequences. This study presents an automated detection of Computed Tomography (CT) kidney stone images using an inductive transfer-based ensemble Deep Neural Network (DNN). Three datasets are created for feature extraction from kidney CT images using pre-trained DNN models. After assembling several pre-trained DNNs, such as DarkNet19, InceptionV3, and ResNet101, the ensemble deep feature vector is created using feature concatenation. The Iterative ReliefF feature selection method is used to choose the most informative ensemble deep feature vectors, which are then fed into the K Nearest Neighbor classifier tuned using a Bayesian optimizer with a 10-fold cross-validation approach to detect kidney stones. The proposed strategy achieves 99.8% and 96.7% accuracy using the quality and noisy image datasets, which are superior to other DNN-based and traditional image detection approaches. This proposed automated approach can help urologists confirm their physical inspection of kidney stones, reducing the possibility of human mistakes.

INDEX TERMS Cross-validation, deep learning, computed tomography, kidney stone, transfer learning, ensemble network.

I. INTRODUCTION

Kidney diseases affect people of all ages and genders. Early detection of kidney diseases is critical, as it is for other diseases. Chronic kidney disease can be deadly if not addressed. Kidney stones must be identified and diagnosed as soon as possible. Early detection of tiny kidney stones helps to avoid the development of chronic kidney disorders. There is a constant increase in kidney patients globally, and several nations (particularly third-world countries) have a shortage of nephrologists [1]. As a result, many people with kidney disease cannot obtain sufficient care. Patients with kidney disorders should be frequently screened, including medical imaging technology screening. These normal processes take time for physicians, and this hectic working schedule may result in an incorrect diagnosis.

Furthermore, many patients suffer from kidney disorders, and a shortage of physicians may cause the treatment to be severely delayed. Computer-assisted medical solutions have been developed to solve these challenges and diagnose kidney disease at an early stage. Automated solutions reduce physician effort and potential human mistakes [2]. It is also always beneficial to achieve accurate outcomes free of subjectivity. As a consequence, the designed system is always precise and sturdy.

Many automated research studies are now being conducted on breast, lung, and heart disorders. However, there has been little research on the recognition of kidney diseases. To identify kidney disorders, ultrasound, Magnetic Resonance Imaging (MRI), and Computed Tomography (CT) are routinely utilized [3]. CT images have generally been utilized

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to identify, quantify, and segment kidney stone disease. The primary goal of automated applications in clinical practice is to create an autonomous model that can identify kidney disorders and assist physicians in accurate treatment.

Detection of medical imaging is a prominent research topic, and numerous autonomous models have been developed to diagnose the disorder reliably. Many Deep Learning (DL) and Machine Learning (ML) models have been utilized to achieve good performance in medical image recognition [4], [5]. Despite using a single model to recognize medical images, some researchers have used ensemble learning to recognize medical images [41], [42], [43]. Therefore, this study proposes a fusion of inductive transfer-based ensemble Deep Neural Networks (DNNs) feature and ML classifier to recognize kidney stones. The inductive transfer-based ensemble DNNs are used to extract the features from the kidney CT images.

In urology, DL and ML-based methods automatically detect ureteral and kidney stones. In this study, a model is proposed for the automated recognition of kidney stones based on a fusion of ensemble DNN along with inductive transfer and ML technique. The proposed technique aims to develop an automated kidney stone recognition model to alleviate insufficient doctor discrepancies. This automated system serves as a resource for nephrologists and radiologists. This study proposes the Fusion of Inductive transfer-based eNsemble Deep netWork features, iterative rELiefF features selector, and machine Learning (FINDWELL) model for kidney stone detection. Before feature extraction, dataset images are pre-processed in various ways, like image augmentation and brightness enhancement, using the proposed fuzzy inference system. Training DNNs from the beginning necessitates a huge number of labeled images, which is extremely problematic in medical imaging because of the personnel costs and time required to build skillfully classified datasets. To adjust the weights associated with DNNs for medical image recognition, inductive transfer or transfer learning [6] has been used by previous researchers. In inductive transfer, the knowledge of an already trained deep learning model with large datasets is transferred to a new relatively small but similar dataset [7]. All extracted features may not help recognize kidney stone images. Thus, an Iterative ReliefF (IRF) features selector [8] is incorporated in this study to select only the meaningful features to reduce future computational complexity. To demonstrate the success of the ensemble feature, a K-Nearest Neighbor (KNN) classifier tuned by Bayesian Optimizer (BO) [9] is used to recognize the kidney stone images. The efficiency of the proposed FINDWELL model is demonstrated using three datasets, which consist of high-quality and noisy kidney CT images. Accuracy, precision, recall, F1 score, kappa score, Matthews Correlation Coefficient (MCC), Good Detection rate (GDR) and Classification Success Index (CSI) are used to evaluate the performance.

The overall contributions of the FINDWELL model are as follows:

A. PRE-PROCESSING OF THE DATASET IMAGES

1) IMAGE BRIGHTNESS ENHANCEMENT BY USING THE PROPOSED FUZZY INFERENCE SYSTEM

The goal of creating a new Fuzzy Inference System (FIS) is to improve the brightness/contrast of the dataset images, thereby overcoming the limitations of previous techniques. The discrete pixel intensity is fuzzified using input Membership Functions (MFs) and then mapped to the output using IF-THEN rules to generate FIS. Finally, the output MFs are used to generate a defuzzified value.

2) IMAGE AUGMENTATION

Augmentation is used to increase the number of image samples in the dataset and add some variety. FINDWELL model's performance and results can be improved by data augmentation. The data augmentation tools enhance and enrich the data, allowing the model to perform better and more correctly. Data augmentation approaches lower operating expenses by including transformation into datasets. In this study, augmentation is done by using horizontal flip, vertical flip, horizontal+vertical flip, and adding salt and pepper noise in the dataset images so that the FINDWELL model becomes robust from these variations. Three different datasets are generated using the augmentation above approaches to check the method's efficacy.

B. PROPOSED ENSEMBLE DEEP FEATURES BASED ON INDUCTIVE TRANSFER MODELS

This study proposes an Inductive transfer-based new ensemble deep feature extractor, which combines the features extracted from Darknet19, InceptionV3, and ResNet101. As a feature generation function, pre-trained DarkNet19 is chosen to extract discriminative features. DarkNets [26] are lightweight CNNs that are exceptionally effective. For these reasons, DarkNet19 is used to extract the features in this study. InceptionV3 [27] is less computationally costly. As regularizers, it employs auxiliary classifiers. An auxiliary classifier is intended to improve the convergence of very deep neural networks. In very deep networks, the auxiliary classifier is primarily employed to tackle the vanishing gradient problem. In the early rounds of training, the auxiliary classifiers did not result in any improvement. However, in the end, the network with auxiliary classifiers outperformed the network without auxiliary classifiers. To minimize the grid size of feature maps, max pooling, and average pooling were traditionally utilized. The activation dimension of the network filters is enhanced in the InceptionV3 model to minimize grid size effectively. For these reasons, InceptionV3 is used in this study as the second feature extractor. Networks with many layers (even thousands) may be readily trained with ResNet101 [28] without raising the training error percentage. ResNet101 uses identity mapping to assist in solving the vanishing gradient problem. Thus, this model is the 3rd choice in this study for extracting the dataset image feature. Lastly, an ensemble feature vector is developed by concatenating

the features extracted from Darknet19, InceptionV3, and Resnet101.

C. PROPOSED FEATURE SELECTOR

Feature selection is critical in data mining and DL, particularly for high-dimensional data. IRF [8] is a non-parametric feature selection strategy that aims to maximize the accuracy of the recognition algorithms. The IRF feature selector selects meaningful features from the ensemble deep feature set for these reasons.

D. PROPOSED CLASSIFIER

In this study, the KNN classifier is used for kidney stone recognition purposes for the following reasons. This classifier is simple and easy to understand. It is non-parametric. No training is required when using the classifier. This classifier can handle large datasets. BO is used to tune the KNN classifier's hyperparameters.

The rest of the paper is organized as follows. Section II discusses the previous works based on different ML algorithms and inductive transfer-based DNNs for kidney stone detection. Section III discusses the proposed technique's methodology, which covers image augmentation, image enhancement using the proposed fuzzy brightness enhancement technique and generation of three different datasets from the original dataset, feature extraction using ensemble inductive transfer enabled DNNs, feature selection by using IRF and classification using KNN tuned by BO. Section IV is concerned with the results of the FINDWELL model and analysis of the model with the techniques used by previous researchers in kidney stone recognition. Section V deals with the discussion of the article. Finally, Section VI concludes the article.

II. RELATED WORKS

This section discusses studies on different ML techniques, DNN models, and inductive transfer for kidney stone recognition.

Serrat et al. [10] used random forests to recognize kidney stones. The authors present findings from a study of 454 CT kidney stone images. The authors of [11] provide two supervised learning approaches for automating and improving the categorization of kidney stones using a collection of 125 ureteroscope-captured kidney stone images. Image cues that urologists use visually to determine the kidney stone are analyzed and encoded as vectors in the approaches. The feature vectors are then classified using Random Forest (RF) and ensemble KNN classifiers. Verma et al. [12] begin by enhancing the kidney stone images with the median filter, the Gaussian filter, and un-sharp masking. After that, the authors employ morphological procedures such as erosion and dilation to determine the region of interest, and ultimately, the authors apply KNN and Support Vector Machine (SVM) classification approaches to analyze kidney stone images. In [13], authors used AdaBoost to classify 211 kidney stones and 201 phleboliths CT images. The 43 individuals in the independent testing group had radiomics characteristics of 24 kidney stones and 23 phleboliths. For the categorization of 221 x-ray kidney stone images, Aksakalli et al. [14] used several machine learning approaches such as Decision Trees (DT), RF, SVM, Multilayer Perceptron (MLP), KNN, NaiveBayes (BernoulliNB), and DNNs utilizing Convolutional Neural Network (CNN). The DT Classifier produced the best classification results in the trials. There are some significant limitations to choosing only ML algorithms for the recognition of kidney stone images properly. ML requires enormous, inclusive/unbiased, and high-quality data sets to train the model. They may also have to wait for fresh data to be created at times. ML requires adequate time to let the algorithms learn and mature sufficiently to perform their function with high accuracy and relevance. It also needs vast resources to function. This may need more excellent computing resources. Another significant problem is the capacity to understand the outcomes of the algorithms appropriately. Researchers must also carefully select the algorithms for the application. The error made in the early phases is massive, and if not remedied at that time, it causes disaster. Bias and wrongdoing must be dealt with separately; they are unrelated. ML is dependent on two factors: data and algorithms. The two variables determine all of the mistakes. Any errors in any variables would have a significant impact on the outcome.

Because of the limitations mentioned above in ML techniques, nowadays, researchers mainly use DL based techniques to recognize kidney stone images. For the classification of 1799 CT scan kidney stone images, Manoj et al. [15] used VGG16 architecture. VGG16 consists of 16 layers with learnable weights: 13 convolutional layers and three fully connected layers. In [16], authors used ResNet50 to recognize 2959 CT kidney stone images. ResNet is an abbreviation for Residual Network. Microsoft Research launched The ResNet design in 2015, and is widely regarded as one of the most popular Convolutional Neural Network architectures. ResNet-50 is a convolutional neural network of 50 layers (48 convolutional layers, one MaxPool layer, and one average pool layer). For the segmentation of 260 kidney stone CT images, Li et al. [17] used SegNet. SegNet is a model for semantic segmentation. This core trainable segmentation architecture is made up of an encoder network, a decoder network, and a pixel-wise classification layer. The encoder network's design is topologically identical to the VGG16 network's 13 convolutional layers. In [18], authors used a Deep Kronecker Neural network (DKN) to recognize kidney stone CT images. The image input layer, Kronecker convolutional layer (Conv2D), Rectifier Linear unit (ReLu) activation layer, max pooling layer, flatten layer, Dropout layer, and Dense layers are all combined in the developed model. Inception-V3 architecture with 48 layers is used by Sabuncu et al. [19] to recognize 8209 kidney stone CT images. In [20], authors used the Deep Believe Network (DBN) to detect 1377 kidney stone CT images. DBNs have no direction in the first two levels, but the layers above them contain directed linkages to lower layers. DBNs are distinct

from standard neural networks in that they may function as generating and discriminative models. DBNs are also distinct from other deep learning algorithms, such as Restricted Boltzmann Machines (RBMs) or autoencoders, because they do not deal with raw inputs like RBMs. Instead, they start with an input layer with one neuron per input vector and go through several layers until they reach a final layer where outputs are created using probabilities derived from the activations of previous levels. To recognize 150 kidney stone CT images, Sesha et al. [21] used AlexNet, an 8-layered DNN architecture. DenseNet169, MobileNetV2, and GoogLeNet are used by Bhardwaj and Shreenidhi [22] to recognize the same. Compared to AlexNet, the GoogLeNet [23] design significantly decreases the number of parameters learned. The architecture comprises 22 layers, including convolutional, activation, and maximum pooling layers. MobileNetv2 [24] was built with 53 layers, and roughly 3.5 million parameters were learned. To extract features from images, MobileNetV2 employs lightweight depth-wise convolutions. DenseNet169 [25] has 169 layers and is relatively low in parameters compared to other models. It also handles the vanishing gradient problem effectively. The authors employed the cross-residual network (XResNet-50) model in [40] to identify kidney stones. The XResnet-50 architecture is divided into four phases. In the Stem and MaxPooling layers, the picture resolution is lowered by half. Furthermore, each level contains ResLayer blocks that degrade the resolution by half, and these blocks are made up of many layers. The XResNet-50 deep model was trained from scratch on raw kidney CT images. This investigation included 433 patients, 278 stone positive and 165 normal. Although these DNNs are incredibly precise, they require many weeks of training, making them computationally challenging.

The majority of available approaches identify kidney stones using a single model. However, the proposed approach (FINDWELL) combines the features extracted from three independent DNN models (ensemble model) and selected using IRF feature selector in fusion with the KNN classifier tuned by BO, resulting in a more efficient detection model than the distinct models. Ensemble learning is the process of combining numerous deep learning models. This is done to improve deep learning model predictions, classification, or other functions. Ensemble learning may also be used to develop a new model by combining the functionality of many deep learning models. Creating a new model offers several advantages over training a new model from the start. Here are a few examples. Because most learning is obtained from coupled models, the new ensemble model takes comparatively less data to train. New ensemble models offer more accuracy and capabilities than those used to create the new model as features are learned by several models. Thus, the advantages and the feature learning capabilities of different deep models can be used in ensemble models. The presented approach performs effectively in both quality and noisy images. The FINDWELL model produced the most significant classification accuracy compared to state-of-the-art approaches.

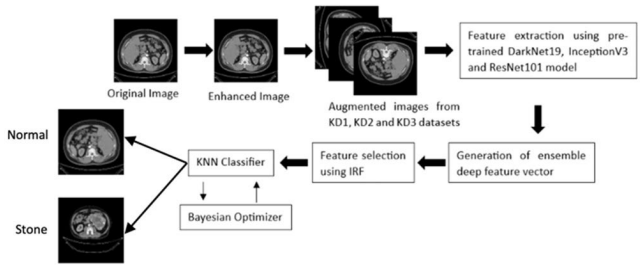


FIGURE 1. The block diagram of the proposed FINDWELL model.

III. PROPOSED METHODOLOGY

The proposed methodology section is divided into five sub-sections, including the proposed image fuzzy brightness enhancement technique, the creation of three different datasets (KD1, KD2, and KD3) by using image augmentation technique, feature extraction using inductive transfer-based DNNs, and the creation of ensemble feature vector, feature selection by using IRF feature selection function, the classification using KNN classifier and the performance measure metric. The proposed approach (FINDWELL) is based on Pre-Trained DNN (PT-DNN) models. Training the DNN models from the beginning necessitates a large number of labeled training data, which is highly problematic in detecting medical images, which has been solved by employing inductive transfer and image augmentation approaches.

Figure 1 shows the block diagram of the FINDWELL model.

A. IMAGE FUZZY BRIGHTNESS ENHANCEMENT

This part aims to provide an innovative FIS to improve image brightness while avoiding the disadvantages of previous techniques [29]. The grey-level image intensities are projected onto a fuzzy space using MFs, the MFs are modified for contrast improvement, and the fuzzy space is mapped back to the grey-level image intensities.

The input CT image (of size $\boldsymbol{Q} \times \boldsymbol{S}$) color mode is first converted to CIELab from the RGB color mode, and the L channel is transformed [39]. The L channel in CIELab color mode displays brightness and depicts a human's eye sensitivity to light during daylight circumstances. As a result, the overall visual contrast may be improved by adjusting the L channel.

After that, the average pixel intensity ($\boldsymbol{P_A}$) is computed using equation (1), where $\boldsymbol{I_{QS}}$ is the pixel intensity at position $\boldsymbol{QS}$.

$$P_A = \frac{\sum_Q \sum_S I_{QS}}{Q \times S} \tag{1}$$

The $\boldsymbol{L}$ channel is altered by utilizing eight MFs (equations (2) - (9)) that vary with $\boldsymbol{I}$ and $\boldsymbol{P_A}$.

$$\mu_{\text{Extremely}Dark} = e^{-(I-(-25))^2 \big/ 2\times\left(\frac{P_A}{9}\right)^2} \tag{2}$$

$$\mu_{VeryDark} = e^{-(I)^2 / 2\times\left(\frac{P_A}{9}\right)^2} \tag{3}$$

$$\mu_{Dark} = e^{-\left(I-\frac{P_A}{2}\right)^2 / 2\times\left(\frac{P_A}{9}\right)^2} \tag{4}$$

$$\mu_{MediumDark} = e^{-\left(I-\left(5\times\frac{P_A}{4}\right)\right)^2 / 2\times\left(\frac{P_A}{9}\right)^2} \tag{5}$$

$$\mu_{MediumBright} = e^{-\left(I-\left(P_A+\frac{255-P_A}{9}\right)\right)^2 / 2\times\left(\frac{255-P_A}{9}\right)^2} \tag{6}$$

$$\mu_{Bright} = e^{-\left(I-\left(P_A+\frac{255-P_A}{2}\right)\right)^2 / 2\times\left(\frac{255-P_A}{9}\right)^2} \tag{7}$$

$$\mu_{VeryBright} = e^{-(I-255)^2 / 2\times\left(\frac{255-P_A}{9}\right)^2} \tag{8}$$

$$\mu_{ExtremelyBright} = e^{-(I-300)^2 / 2\times\left(\frac{255-P_A}{9}\right)^2} \tag{9}$$

In this study, bright pixels are converted to brighter and dark pixels to boost the brightness of the input image. This technique generates the rulebase for this fuzzy system, as indicated in equation (10). In equation (10), OL and IL denote the output and input L channel pixel values, respectively.

$$\begin{aligned} IL_{VeryBright} &\rightarrow OL_{ExtremelyBright} \\ IL_{Bright} &\rightarrow OL_{VeryBright} \\ IL_{MediumBright} &\rightarrow OL_{Bright} \\ IL_{MediumDark} &\rightarrow OL_{Dark} \\ IL_{Dark} &\rightarrow OL_{VeryDark} \\ IL_{VeryDark} &\rightarrow OL_{ExtremelyDark} \end{aligned} \tag{10}$$

Once the L channel membership value is determined, the updated L channel value (L_{mod}) is computed using equation (11).

$$L_{mod} = \frac{\sum_{i=1}^{Q} I_i \cdot \mu(I_i)}{\sum_{i=1}^{Q} \mu(I_i)} \tag{11}$$

The $\boldsymbol{L}$ channel might be enhanced by substituting the original $\boldsymbol{L}$ values with the $\boldsymbol{L_{mod}}$ values, resulting in a better $\boldsymbol{L_{mod}}$ channel. This modified detail may be coupled with the conserved chromatic data (a and b channels) to produce a modified CIELabE image, which can then be converted into an improved $\boldsymbol{RGB_E}$ image. Using the FIS, enriched samples are created from each kidney CT image.

The Peak Signal to Noise Ratio (PSNR) is determined using equation (12) to evaluate the performance of the proposed FIS-based image enhancement approach.

$$PSNR = 10log_{10}\left(\frac{H^2}{RMSE}\right)$$

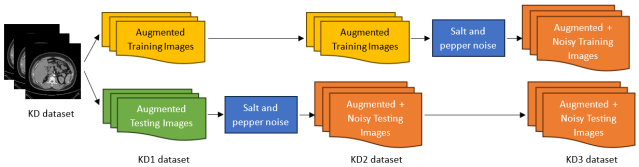


FIGURE 2. Transformation of datasets: from KD to KD3.

$$RMSE = \frac{\sum_{Q}^{S} (RGB(Q,S) - RGB_E(Q,S))^2}{Q \times S} \tag{12}$$

where $\boldsymbol{H}$ denotes maximum kidney CT image intensity, $(\boldsymbol{Q},\boldsymbol{S})$ represents the image size, and RGB and $\boldsymbol{RGB_E}$ denote the original and enhanced images, respectively.

B. IMAGE AUGMENTATION

Image augmentation is the most often utilized strategy for reducing overfitting during the DNN training phase [30]. Image augmentation does not affect the semantic meaning of the images. Training the network on a bigger augmented dataset increases the resilience and generalizability of DNN to new data [31]. To create 6000 normal and 6000 kidney stone CT images, various image augmentation techniques such as horizontal flip, vertical flip, horizontal + vertical flip, and salt and pepper noise are applied.

CT images often contain noise. This is the motivation for adding salt and pepper noise to the dataset. "Salt and pepper noise" refers to a condition in which random pixels are replaced with exceptionally dark or bright values. While doing a CT scan of the kidney, the same situation can occur where due to sharp and sudden disturbances in the image signal, the scan is having the impact of salt and pepper noise.

Three kidney CT image dataset variants, i.e., KD1, KD2, and KD3, are created from the original KD dataset. KD1 is created by applying horizontal flipping, vertical flipping, and horizontal + vertical flipping on the images of KD. On the other hand, noisy kidney CT images can be found in the KD2 dataset which are generated by applying salt and pepper noise on the KD1 images. In KD2, the noise is applied only to the testing images. KD3 is generated using KD2 images where the salt and pepper noise is applied to the training set CT images.

The transformation of the dataset images from KD to KD3 is shown in Figure 2. The datasets KD2 and KD3 are created to test the FINDWELL model's performance in a noisy environment.

C. FEATURE EXTRACTION USING INDUCTIVE TRANSFER-BASED ENSEMBLE DNNs

Inductive transfer is the technique of training a network on a large dataset of annotated images to acquire the knowledge of general image descriptors or features and then transferring the retrieved features to a relatively small dataset for detection [6]. Figure 3 depicts the inductive transfer framework for PT-DNN. In this study, PT-DNN models are used that were created by training DNN architectures from the beginning

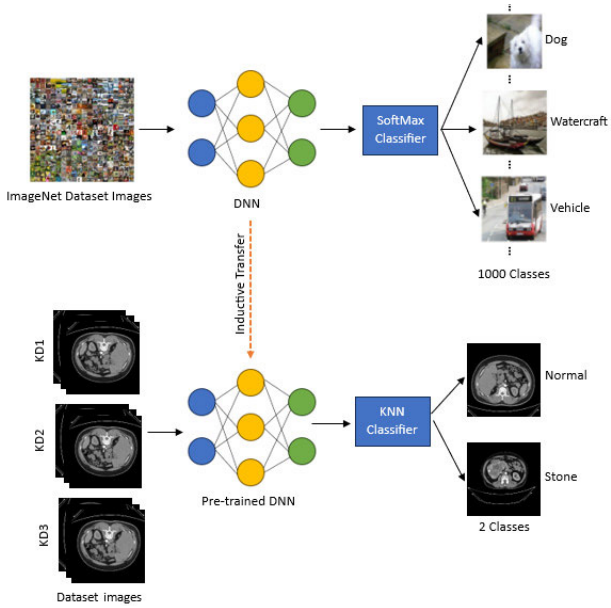


FIGURE 3. The inductive transfer framework of PT-DNN for kidney stone detection.

with images obtained from the ImageNet database [32]. 1.2 million images are used to train the model to recognize 1000 categories. During the training phase, the SoftMax classifier receives features derived by the DNN for categorizing 1000 class images, as illustrated in Figure 3. Three PT-DNNs, DarkNet19 [26], InceptionV3 [27], and ResNet101 [28] are employed in the proposed method to detect CT kidney stone images automatically.

Each PT-DNN model is trained and tested using several types of datasets KD1, KD2, and KD3. These datasets are created by adding some variations in the dataset images. The total number of kidney CT images in the dataset is 12000 (Normal: 6000, Stone: 6000). The PT-DNN model trained on ImageNet dataset images supports learning descriptors or features from kidney CT images. Because the FINDWELL model is built from an ensemble of multiple PT-DNNs, the feature vector from each model is combined to produce the final anticipated feature vector for further processing.

To decide which DNNs are to be combined to generate the ensemble DNN, the experiment is conducted by using the KD1 dataset as it contains quality images and the dataset is balanced. To improve the classification model, the three best performances of PT-DNNs are examined and ensembled together. The goal of the study is to create a lightweight, computationally less expensive ensemble model.

These PT-DNNs are trained on kidney CT images to detect kidney stone images. The ensemble method's primary goal is to integrate the predictions of various models to get superior classification output [33]. The ensemble approach can potentially minimize network model prediction variation [34] and bias [34]. As a result, three PT-DNN models are assembled in the proposed method for reliable detection of CT kidney stone images. Figure 4 depicts the ensemble feature vector generation technique for the proposed strategy.

FIGURE 4. Ensemble feature vector creation for the FINDWELL model.

The feature vectors generated from PT-DNN 1, 2, and 3 are concatenated to generate the ensemble feature vector.

D. FEATURE SELECTION FROM INDUCTIVE TRANSFER-BASED ENSEMBLE DNNs

Ensemble DNN extracted many features during the feature generation phase, and the most valuable descriptors or features are chosen utilizing an Iterative ReliefF (IRF) features selector [8]. Relief is a frequently used feature selection approach that is feature precision and discovers the estimator's weights when the estimator's output is dependent on a multiclass variable. The relief algorithm is used to rank features for the sample chosen from the data set, considering the proximity of other classes in its class and its distance from other classes. Once this feature ranking is done using a model with negative and positive weights, the feature selection procedure is finished. To the best of our knowledge, the Relief algorithm and its variations are the only individual evaluation filter methods capable of identifying feature dependencies. These algorithms do not search through feature combinations but rather employ the idea of closest neighbors to obtain feature statistics that account for interactions in an indirect manner. To improve the classification ability, the feature reduction strategy is applied.

ReliefF creates positive and negative weights and calculates them using the Manhattan distance. ReliefF is an improved version of Relief and chooses the most noticeable aspects. The relief-based feature selection approach employs Euclidean distance. The ReliefF weights are built using the Manhattan distance. As a result, the IRF approach is suggested.

In IRF, the previous iteration's feature weights W are used to update pairwise distance computations each time, such that a low-scoring feature from the prior iteration has less effect on instance distance in the current iteration. When the distance weights are iteratively updated, certain samples may enter and leave the neighborhoods of other samples. Iterative Relief additionally included a radius to define neighborhoods rather than a predetermined number of instances to decrease discontinuities in feature weight estimations caused by shifting neighborhoods. Iterations were repeated until the weights converged or a maximum number of iterations was achieved.

The main motivation for using the IRF method is the iteration range reduces the time, and the loss is calculated at every feature selection. Also, this method is less sensitive to noise as it looks for n nearest neighbors iteratively. This method can also handle the missing values. The IRF feature selector selects the most valuable features by iteratively removing the

irrelevant features, which are used for the following processing. For optimal qualities, a loss calculator should be used. As a result, in this study, the loss value calculator is a KNN classification algorithm tuned by using BO with a 10-fold Cross-Validation (CV).

E. KIDNEY STONE CT IMAGES CLASSIFICATION BY USING KNN OPTIMIZED BY BAYESIAN OPTIMIZER

The ensemble deep features are classified using a KNN classifier. The Bayesian Optimization (BO) [9] method is used to adaptively optimize the hyperparameters of KNN to increase prediction performance. The critical advantage of BO is that it is resistant to non-convex issues and is less prone to slip into a local optimum. The BO procedure requires a prior function, such as the frequently used Gaussian kernel, to optimize the objective function. The latter is defined by its mean function and covariance function, both of which are computed from data points. The kind of distance metric (Chebychev, cityblock, cosine, correlation, Hamming, Euclidean, Mahalanobis, Jaccard, Seuclidean, Minkowski, Spearman) and the number of neighbors given by K are the parameters of KNN to be optimized in this study. BO's fitness function produced the lowest misclassification rate.

F. PERFORMANCE MEASURE

FINDWELL model's quantitative evaluation is based on precision, recall, accuracy, F1 score, kappa score, MCC, GDR and CSI measures. The values TP, FP, TN, and FN reflect the expected true positive, false positive, true negative, and false negative samples, respectively. Equations 13 – 20 provide the performance measures.

$$Precision = \frac{TP}{TP + FP} \tag{13}$$

$$Recall = \frac{TP}{TP + FN} \tag{14}$$

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN} \tag{15}$$

$$F1Score = \frac{2 \times Precision \times Recall}{Precision + Recall} \tag{16}$$

$$Kappa = \frac{2 \times (TP \times TN - FN \times FP)}{(TP + FP) \times (FP + TN) + (TP + FN) \times (FN + TN)} \tag{17}$$

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP) \times (TP + FN) \times (TN + FP) \times (TN + FN)}} \tag{18}$$

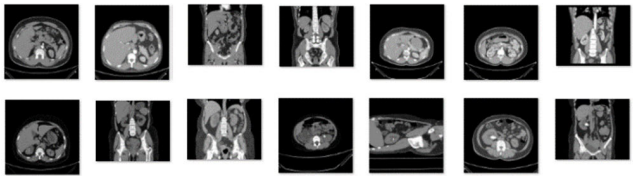


FIGURE 5. Samples from the KD dataset: normal (first row), kidney stone (last row).

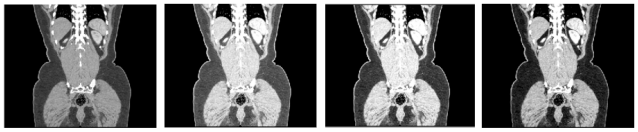


FIGURE 6. Image enhancement using fuzzy logic: (First) Original image, (Second) Enhancement using HE, (Third) Enhancement using CLAHE, (Last) Enhancement using the proposed approach.

$$GDR = \frac{TP - FP}{TP + FN} \tag{19}$$

$$CSI = \frac{TP}{TP + FP} + \frac{TP}{TP + FN} - 1 \tag{20}$$

These performance matrices are utilized to compute the model's performance in classifying kidney CT images. The performance measurements are computed after training and testing with different i.e., KD, KD1, KD2, and KD3 datasets.

IV. RESULTS AND ANALYSIS

For the training of the FINDWELL model, Intel Core i7, 2.6 GHz 6-Core processor, NVIDIA GeForce GTX 1600 Ti Graphics (6GB), and 16 GB 2667 MHz DDR4 RAM, is used which minimizes the time of training. TensorFlow 2.9.2 is used for the implementation of the suggested approach.

A. DATASETS

The kidney CT images are collected from [35]. The dataset was gathered through PACS (Picture archiving and communication system) from several hospitals in Dhaka, Bangladesh, where patients had already been diagnosed with normal or stone kidney results. The Coronal and Axial slices were chosen from both contrast and non-contrast examinations, and the entire abdomen and urogram protocol was followed. The Dicom study was then carefully picked, one diagnosis at a time, and a batch of Dicom images of the region of interest for each radiological result was made from those. Each patient's information and metadata are then removed from the Dicom images and converted to a lossless jpg image format. Following the conversion, each imaging finding was double-checked by a radiologist and a medical technician to ensure that the data was correct. The collected dataset consists of 5077 normal and 1377 kidney stone images. In this study, this dataset is named as KD dataset. Figure 5 represents some sample images from the KD dataset.

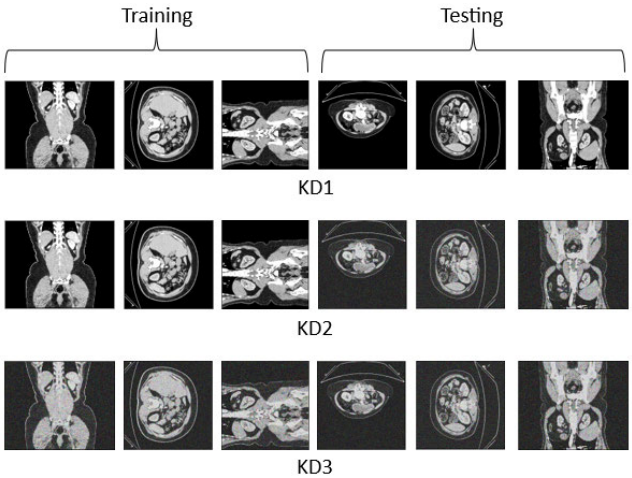


FIGURE 7. Samples from KD1 (Training: Augmented, Testing: Augmented), KD2 (Training: Augmented, Testing: Augmented + Noisy), KD3 (Training: Augmented + Noisy, Testing: Augmented + Noisy) datasets. The first three images are from the training set and the last three images are from the testing set.

TABLE 1. The proportion of data for training, validation, and testing.

Dataset	Set	No. of images	% of images
KD	Training	4647 (N: 3656, S:991)	72
	Validation	516 (N: 406, S: 110)	8
	Testing	1291 (N: 1015, S:276)	20
	Total	6454 (N: 5077, S: 1377)	100
KD1, KD2 and KD3	Training	8640 (N: 4320, S: 4320)	72
	Validation	960 (N: 480, S: 480)	8
	Testing	2400 (N: 1200, S: 1200)	20
	Total	12000 (N:6000, S: 6000)	100

*N: Normal, S: Stone

B. IMAGE BRIGHTNESS ENHANCEMENT BY USING THE PROPOSED APPROACH

After collecting the dataset images, the first step is to enhance the brightness of the KD dataset images using the proposed FIS. The outcome of the proposed fuzzy image brightness enhancement approach is compared to that of two well-known image enrichment methods, histogram equalization (HE) and Contrast-Limited Adaptive Histogram Equalization (CLAHE). The results of the HE, CLAHE, and suggested fuzzy image brightness enhancement approaches are shown in Figure 6.

PSNR is used to evaluate the efficacy of the proposed fuzzy image brightness enhancement method. CLAHE, HE, and the proposed FIS enrichment technique have average PSNRs of 9.5, 8.2, and 10.7, respectively.

C. KD1, KD2 AND KD3 DATASETS FROM KD

As the KD dataset is not balanced, an image augmentation technique is employed on KD dataset and three different datasets are created: KD1, KD2, and KD3 which are balanced and having different varieties. After augmentation, each dataset contains 6000 normal images and 6000 kidney stone CT images.

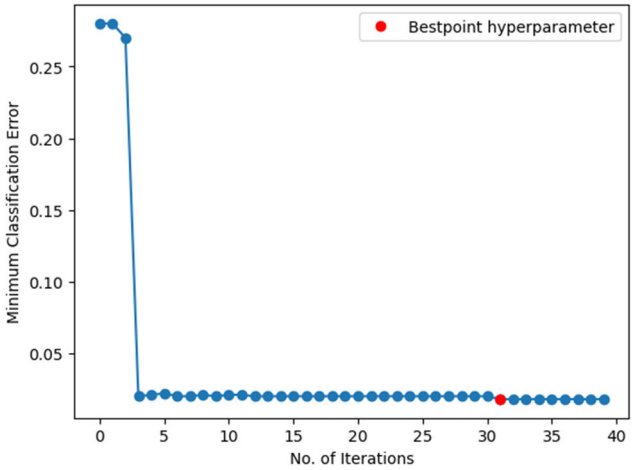


FIGURE 8. The best point hyperparameter of the utilized KNN classifier.

TABLE 2. Network analysis of several PT-DNNs.

DNN	Layer depth	Input image size	Learnable parameters
AlexNet [21]	8	227 x 227 x 3	60 million
GoogleNet [22]	22	227 x 227 x 3	6.4 million
DenseNet169 [22]	169	224 x 224 x 3	141 million
MobileNetV2[22]	105	224 x 224 x 3	3.5 million
ResNet101[28]	209	224 x 224 x 3	44.7 million
SqueezeNet [36]	18	224 x 224 x 3	3.2 million
InceptionV3 [27]	189	299 x 299 x 3	23.9 million
VGG16 [27]	16	224 x 224 x 3	138.4 million
ShuffleNet [37]	50	224 x 224 x 3	3.4 million
DarkNet19 [26]	19	256 x 256 x 3	88.1 million

TABLE 3. Performance comparison of several PT-DNNs.

DNN	Accuracy (%)	p-value
AlexNet [21]	88.4	$3.28e^{-310}$
GoogleNet [22]	87.9	$1.76e^{-302}$
DenseNet169 [22]	89.7	0.0
MobileNetV2[22]	91.8	0.0
DarkNet19 [26]	96.3	0.0
InceptionV3 [27]	95.3	0.0
VGG16 [27]	85.4	$5.12e^{-263}$
ResNet101[28]	95.6	0
SqueezeNet [36]	74.5	$8.44e^{-128}$
ShuffleNet [37]	94.1	0

Some samples from KD1, KD2, and KD3 datasets are depicted in Figure 7.

D. TRAINING CRITERIA

We choose 80% of the kidney CT images for training and 20% for testing for the individual and ensemble DNN models. On training images, we employed 10-fold cross-validation. In the training, validation, and testing sets, we have an equal

TABLE 4. Performance comparison of several ensemble DNNs.

Ensemble DNN	Accuracy (%)	p-value
Ensemble 1 (Combination of DarkNet19 [26], ResNet101 [28], InceptionV3 [27], ShuffleNet [37] and MobileNetV2 [22])	90.7	0.0
Ensemble 2 (Combination of DarkNet19 [26], ResNet101 [28], InceptionV3 [27] and ShuffleNet [37])	95.3	0.0
Ensemble 3 (Combination of DarkNet19 [26], ResNet101 [28] and InceptionV3 [27])	97.1	0.0
Ensemble 4 (Combination of DarkNet19 [26] and ResNet101 [28])	96.3	0.0

proportion of normal and kidney stone CT images. Table 1 shows the proportion of data for training, validation, and testing of the individual and proposed ensemble DNN model.

E. SELECTION OF BEST POINT HYPERPARAMETER VALUE USING KNN CLASSIFIER

To find out the best point hyperparameter value using the KNN classifier, 40 iterations are considered. Figure 8 depicts the best point hyperparameter of the utilized KNN classifier. To carry out this experiment the KD1 dataset is used. The optimized K value is 1 with the Euclidean distance metric.

F. SELECTION OF DNNs TO CREATE THE ENSEMBLE DNN FOR FEATURE EXTRACTION

Table 2 shows the network analysis of several PT-DNNs used in this study.

The performance of individual PT-DNNs is compared to create the ensemble DNN using a 10-fold CV which is listed in Table 3. Here the images from the KD1dataset are used as it contains quality images and the dataset is balanced.

From Table 3 we can conclude that the top five PT-DNNs which result in good performances and p-value < 0.05 are DarkNet19 [26], ResNet101 [28], InceptionV3 [27], ShuffleNet [37], and MobileNetV2 [22]. The combination of these PT-DNNs is used to create the proposed ensemble DNN. The performance comparison of the combination of these PT-DNNs is listed in Table 4.

From Table 4 it is clear that Ensemble 3 which is the combination of DarkNet19 [26], ResNet101 [28], and InceptionV3 [27] is producing the best performance (97.1% accuracy) compared to other ensemble DNNs. As a result, DarkNet19 [26], InceptionV3 [27], and ResNet101 [28] are selected for their unique performance and lightweight structure. Darknet19 is faster and more precise since it is less complicated. The advantage of using InceptionV3 is the use of an auxiliary classifier. In very deep networks, the auxiliary classifier is primarily employed to tackle the vanishing gradient problem. In the early rounds of training, the auxiliary classifiers did not result in any improvement. However, in the end, the network with auxiliary classifiers outperformed the network without auxiliary classifiers. InceptionV3 uses label smoothing

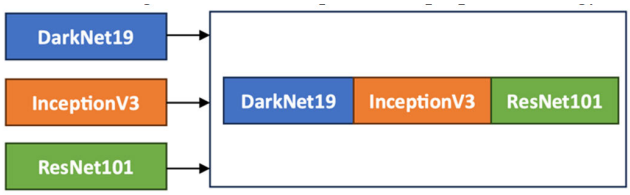


FIGURE 9. Ensemble feature vector creation for the FINDWELL model.

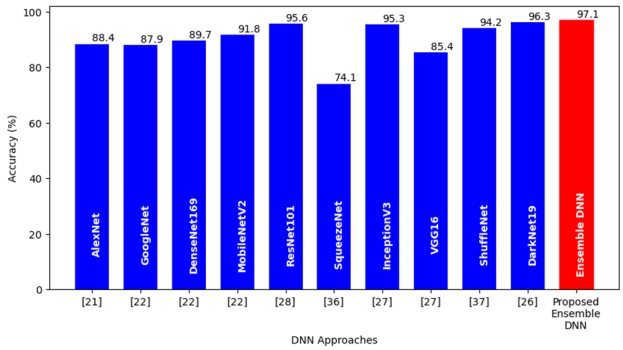


FIGURE 10. Performance comparison of different DNN approaches as well as the proposed ensemble technique.

regularization. It is a technique for regularizing the classifier by assessing the impact of label dropout during training. It stops the classifier from making too confident predictions about a class. ResNet101 provides heuristics to increase the parallelism of training and decrease the computational cost through lower precision computing and modifying the learning rate or biases.

The following are the network parameters of the DNNs used in this study: DarkNet19 [26], InceptionV3 [27], and ResNet101 [28]. DarkNet19 consists of 19 Convolutional Layers, 18 Batch Normalization, 18 Leaky ReLU, five max-pooling, and 1 SoftMax with a weight decay of 0.005 and momentum of 0.9. The learning rate is set to 0.0002 and the batch size is 512. After the training phase, 88.1 million parameters are learned. InceptionV3 is 48 layers deep with a weight decay of 0.9 and momentum of 0.9. The learning rate for the network is set to 0.001, and batch sizes range from 4,096 to 16,384. ResNet101 is 101 layers deep with a weight decay of $10e^{-4}$ and momentum of 0.9. The quickest training progress is seen at a learning rate of 0.1 and a batch size of 768. After the training phase, 23.9 million parameters are learned. ResNet101 is 101 layers deep with a weight decay of 0.0001 and momentum of 0.9. The learning rate and batch size are set to 0.001 and 64. After the training phase, 44.7 million parameters are learned. Adam optimizer is used to train every DNN. Henceforth, the combination (concatenation) of the features generated from these PT-DNN architectures, i.e., DarkNet19, InceptionV3, and ResNet101, are employed in the proposed FINDWELL model. Figure 9 depicts the ensemble feature vector generation technique for the proposed strategy.

The feature vectors generated from DarkNet19, InceptionV3, and ResNet101 are concatenated to generate the ensemble feature vector.

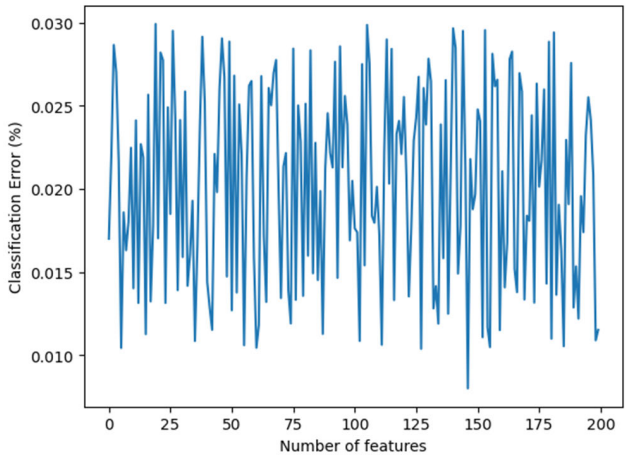


FIGURE 11. Classification error vs the number of features utilized with the IRF features selector.

TABLE 5. The model's accuracy comparison using IRF, TuRF, and VLSReliefF feature selectors on the test samples of the KD1 dataset.

Approaches	Accuracy (%)	Time (in Hrs)
Proposed ensemble DNN + VLSReliefF + KNN	98.4	10
Proposed ensemble DNN + TuRF + KNN	99.0	9
Proposed ensemble DNN + IRF + KNN	99.8	6

Fig. 10 represents the accuracy comparison of the performance among the existing DNNs and the proposed ensemble method.

According to Figure 10, the proposed technique obtains a higher classification accuracy of 97.1% when compared to the individual PT-DNNs [21], [23], [26], [27], [28], [36], [37]. The proposed approach ensembles the prediction findings from DarkNet19 [26], InceptionV3 [27], and ResNet101 [28], resulting in a more powerful classification model. This implies that the proposed ensemble DNN can properly detect CT kidney stone images with fewer misclassification errors.

G. DETECTION OF CT KIDNEY STONE IMAGES BY USING SELECTED DEEP FEATURES USING IRF FROM THE KD1 DATASET

After constructing the DNN ensemble for the CT kidney stone image detection, first, the performance of the proposed DNN ensemble is tested on the KD1 dataset. Performance measures such as accuracy, precision, recall, F1-score, kappa score, MCC, GDR, and CSI are utilized to evaluate the performance. For that, the first task is to select the number of meaningful features by using the IRF feature selector. Number of features generated per image from the proposed ensemble DNN (DarkNet19 (features: 1024), InceptionV3 (features: 2048), and ResNet101 (features: 2048)) is 5120. To select the optimal number of features for the detection of kidney CT images we used the iteration range of {100:300}. Figure 11 depicts a graph of classification error vs the number of features utilized with the IRF features selector. The best feature vector has a length of 146, as shown in Figure 11. KNN classifier tuned using BO is used to classify these deep features (see Figure 8).

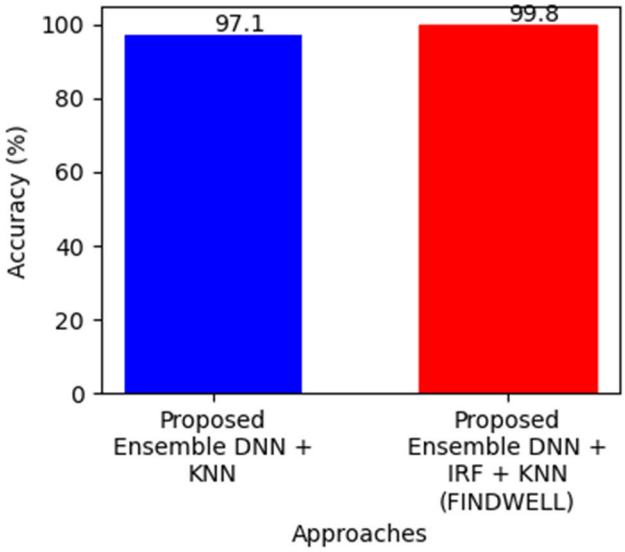


FIGURE 12. The classification performance comparison uses the proposed normal ensemble DNN and FINDWELL model.

TABLE 6. Comparison of accuracy on KD1 dataset using different CV folds.

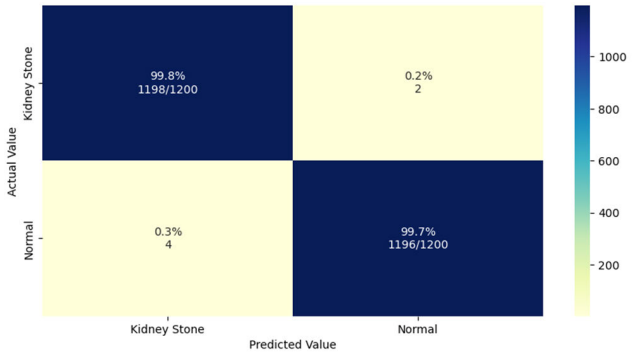
CV	Accuracy (%)
5-fold	98.3
10-fold	99.8
15-fold	96.7
20-fold	96.1
25-fold	94.8
30-fold	94.5

IRF choose 146 optimal features per image from the produced 5120 features, which are then input into the KNN classifier tuned by BO for the classification of CT kidney stone images.

The performance of the proposed ensemble model on KD1 test samples is tested with IRF feature selector and is compared with another two iterative feature selection methods i.e., TuRF and VLSReliefF. Table 5 depicts the model's performance comparison using IRF, TuRF, and VLSReliefF feature selectors.

From Table 5, it can be noticed that there is no significant difference in the accuracy obtained using the proposed ensemble DNN and IRF, TuRF, and VLSReliefF feature selector but we can see a significant difference in the time of execution. As time is also an important factor in any automated detection system, we used the IRF feature selector for further processing.

Fig. 12 shows the performance comparison while using normal ensemble DNN features and ensemble DNN + selected features using IRF for the detection. The proposed FINDWELL model obtained a better classification accuracy of greater than 99% using the IRF-selected features and 10-fold CV.




```
Accuracy   : 0.9975
Precision : 0.9966722129783694
Recall    : 0.9983333333333333
F1-score  : 0.9975020815986677
Kappa score   : 0.995
MCC   : 0.9950013819473235
CSI   : 0.9950055463117027
```

FIGURE 13. Confusion matrix and performance measure generated using the FINDWELL model when applied to the test samples of the KD1 dataset.

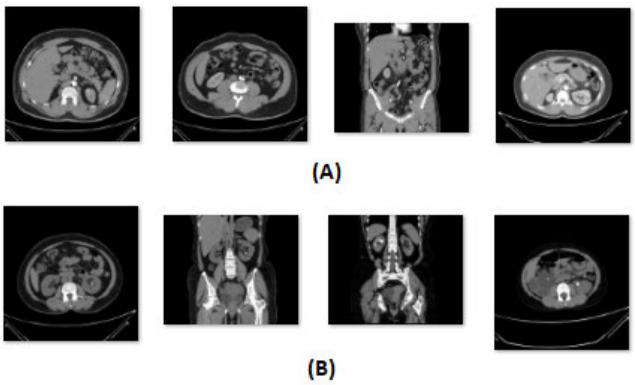


FIGURE 14. Some correctly classified test samples using the FINDWELL model from the KD1 dataset (A) Normal, (B) Kidney stone.

Before calculating all the performance measures using the KD1 dataset, we examined the accuracy values obtained from FINDWELL using different CV folds, i.e., {5, 10, 15, 20, 25, and 30}, which is depicted in Table 6.

Table 6 clearly shows that a 10-fold CV produces better classification results on the KD1 dataset compared to other folds.

Figure 13 depicts the confusion matrix produced using the FINDWELL model with the 10-fold CV on the test samples from the KD1 dataset. Also, the performance measures are included in Figure 13. The confusion matrix provides insight into the number of errors the network model produces when predicting each class.

It is important to note that the FINDWELL model, as illustrated in Figure 13, achieves nearly 100% recall. This demonstrates that the proposed method accurately classifies almost all CT kidney stone images with minor classification mistakes.

Figure 14 depicts some correctly classified test samples using the FINDWELL model from the KD1 dataset.

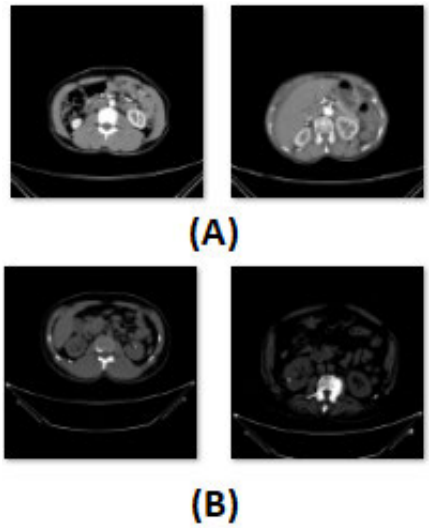


FIGURE 15. Some incorrectly classified test samples using the FINDWELL model from the KD1 dataset (A) Normal, (B) Kidney stone.

Figure 15 depicts some incorrectly classified test samples using the FINDWELL model from the KD1 dataset.

The fundamental issues related to image classification arise from the fact that images are simply massive matrices with a great number of pixels present, and they are extremely complicated. Training a computer to accurately categorize images is a time-consuming and challenging undertaking. In some circumstances, the incorrect features may be utilized to make judgments.

H. PERFORMANCE OF FINDWELL MODEL USING KD2 DATASET AND COMPARISON OF THE PERFORMANCE WITH EXISTING DNNs

The dataset KD2 is used to assess the FINDWELL model's performance in a noisy environment. This dataset KD2 contains augmented training images and augmented + noisy testing images. Images are generally distorted by noise in actual applications; therefore, validating the FINDWELL model's efficiency in noisy conditions is more practical. Figure 16 depicts the FINDWELL model's performance deterioration when tested with noisy images. As seen in Figure 16, the performance of the networks degrades if noise levels increase. Also, as seen in Figures 13 and 17, there is a considerable decrease in DNN classification accuracy as noise variation increases. This leads to the conclusion that all present approaches are noise-sensitive, and their performance suffers significantly as the noise intensity rises. Another reason for the poor performance can be the disparity in the quality of training and testing images. As a result, the performance of DNN is quite bad, which is rectified by training and testing the models with dataset KD3. At noise levels of 0.02, 0.04, 0.06, 0.08, and 0.1, the proposed technique achieves 85%, 79.6%, 71.4%, 64.7%, and 57.1% accuracy, respectively using the KD2 dataset images.

Before calculating all the performance measures using the KD2 dataset, we examined the accuracy values obtained from

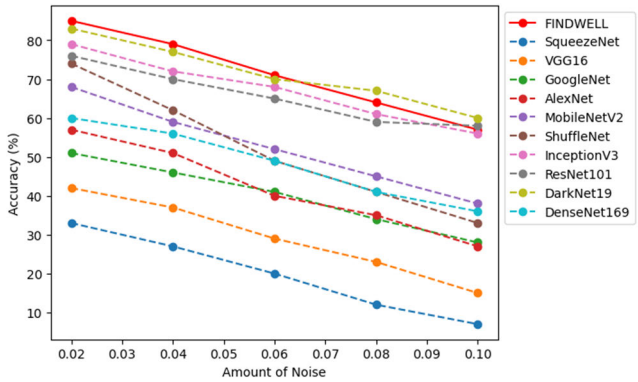
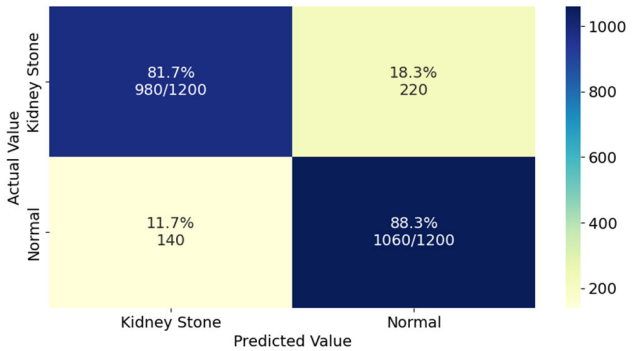


FIGURE 16. Comparison of the performance of the proposed FINDWELL model with existing DNNs using the KD2 dataset.

TABLE 7. Comparison of accuracy on KD2 dataset using different CV folds.

CV	Accuracy (%)
5-fold	80.1
10-fold	85.0
15-fold	83.7
20-fold	83.2
25-fold	81.3
30-fold	78.7




```
Accuracy   : 0.85
Precision  : 0.875
Recall     : 0.8166666666666667
F1-score   : 0.8448275862068966
Kappa score   : 0.7
MCC   : 0.701560760020114
CSI   : 0.6916666666666667
```

FIGURE 17. Confusion matrix and performance measure using the FINDWELL model when applied to the test samples of the KD2 dataset.

FINDWELL using different CV folds i.e., {5, 10, 15, 20, 25, and 30} which is depicted in Table 7.

Table 7 clearly shows that a 10-fold CV produces better classification results on the KD2 dataset compared to other folds.

Figure 17 depicts the confusion matrix produced using the FINDWELL model applied to the KD2 test dataset and the performance measure. This matrix is generated by considering the noise level of 0.02.

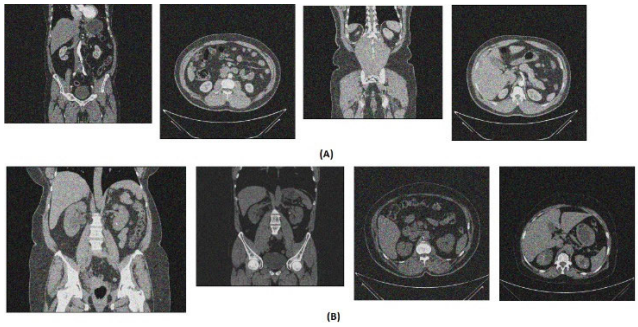


FIGURE 18. Some correctly classified test samples using the FINDWELL model from the KD2 dataset (A) Normal, (B) Kidney stone.

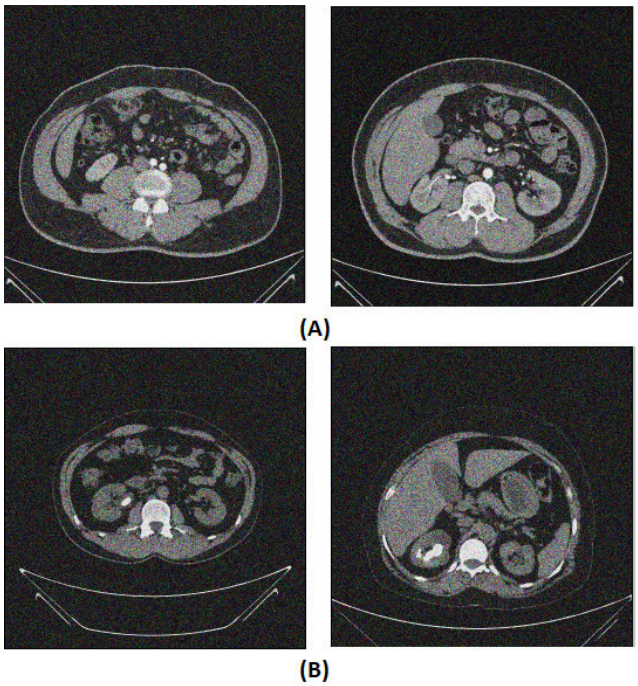


FIGURE 19. Some incorrectly classified test samples using the FINDWELL model from the KD2 dataset (A) Normal, (B) Kidney stone.

From the confusion matrix and the performance measure shown in Figure 17, it can be concluded that while using a DNN-based technique for supervised classification, the training and testing samples should contain similar augmentations. In the KD2 dataset, as the only testing data is noisy, the existing DNN and the proposed technique cannot handle it properly.

Fig. 18 depicts some correctly classified test samples using the FINDWELL model from the KD2 dataset.

Fig. 19 depicts some incorrectly classified test samples using the FINDWELL model from the KD2 dataset.

The main reason for the poor performance of the FINDWELL model using KD2 dataset images is the variation in training and testing image samples. The model is trained with quality images and we've tested the model with noisy images. Because of the presence of the noise in the test samples, the extracted feature vectors are significantly different from the extracted features from the training samples. Thus, the

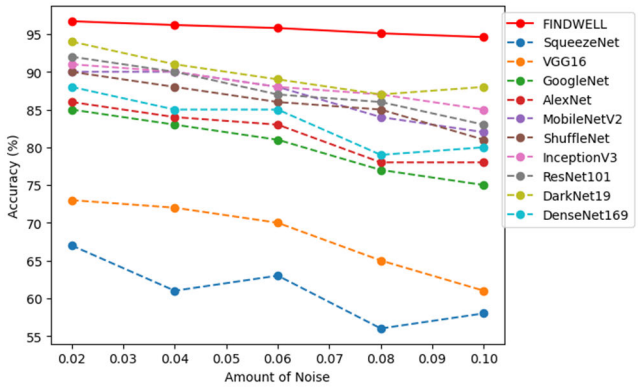


FIGURE 20. Comparison of the performance of the proposed FINDWELL model with existing DNNs using the KD3 dataset.

TABLE 8. Comparison of accuracy on KD3 dataset using different CV folds.

CV	Accuracy (%)
5-fold	96.1
10-fold	96.7
15-fold	95.3
20-fold	94.8
25-fold	94.1
30-fold	92.6

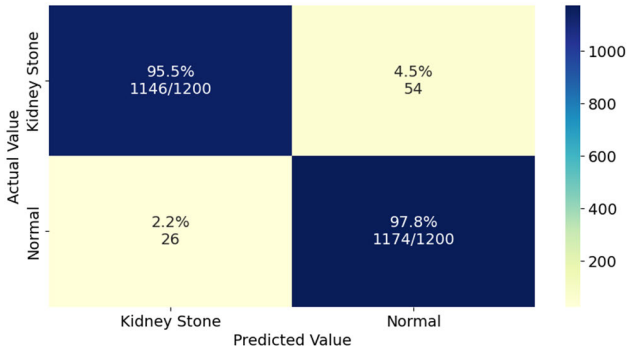
model fails to detect the images correctly even if the main foreground is large and has enough information for the proper image detection. This issue is solved by creating the KD3 dataset.

I. PERFORMANCE OF FINDWELL MODEL USING KD3 DATASET AND COMPARISON OF THE PERFORMANCE WITH EXISTING DNNs

KD3 is a dataset made up of augmented as well as noisy training and testing CT images. The classification performance of the DNN is enhanced by training it with noisy images, as seen in Figure 20. Figures 16 and 20 show that the FINDWELL model performs better at all noise levels when trained with KD3 than with the KD2 dataset. This is mainly because the quality of training and testing is similar to dataset KD3. When compared to previous approaches, Figure 15 clearly shows that the FINDWELL model attains the highest classification accuracy at every noise level. At noise levels of 0.02, 0.04, 0.06, 0.08, and 0.1, the proposed technique achieves 96.7%, 96.2%, 95.8%, 95.1%, and 94.6% accuracy, respectively. This demonstrates that the proposed technique (FINDWELL) can properly categorize images even in the existence of high and low noise levels in the test images. Furthermore, as shown in Figures 13 and 21, the ensemble-based system outperforms other methods in accurately detecting CT kidney images in both high-quality and noisy environments.

Before calculating all the performance measures using the KD3 dataset, we examined the accuracy values obtained from FINDWELL using different CV folds, i.e., {5, 10, 15, 20, 25, and 30}, which is depicted in Table 8.

Table 8 clearly shows that a 10-fold CV produces better classification results on the KD1 dataset compared to other folds.




```
Accuracy   : 0.9666666666666667
Precision  : 0.9778156996587031
Recall     : 0.955
F1-score   : 0.966273187183811
Kappa score   : 0.9333333333333333
MCC   : 0.9335875112014138
CSI   : 0.9328156996587031
```

FIGURE 21. Confusion matrix and performance measure using the FINDWELL model when applied to the test samples of the KD3 dataset.

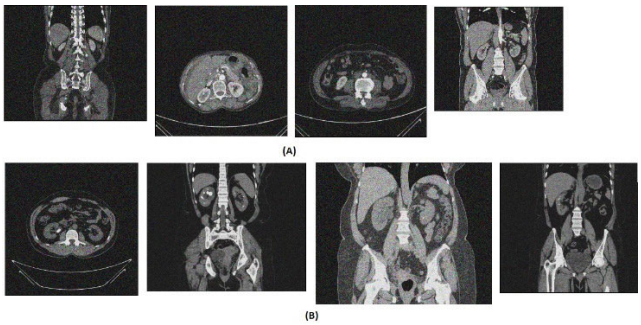


FIGURE 22. Some correctly classified test samples using the FINDWELL model from the KD3 dataset (A) Normal, (B) Kidney stone.

Fig. 21 depicts the confusion matrix produced using the FINDWELL model applied to the test samples of the KD3 dataset as well as the performance measure. This matrix is generated by considering the noise level of 0.02.

One can notice from Figure 21 that the FINDWELL model can efficiently classify the images in a noisy environment.

Fig. 22 depicts some correctly classified test samples using the FINDWELL model from the KD3 dataset.

Fig. 23 depicts some incorrectly classified test samples using the FINDWELL model from the KD3 dataset.

The main reason behind the misclassification of some of the test samples from the KD3 dataset is the presence of noise as the presence of noise can significantly change image features.

J. PERFORMANCE OF FINDWELL MODEL ON KD DATASET IMAGES

Lastly, we tested the performance of the proposed method on the KD dataset, which contains original (without augmentation) images.

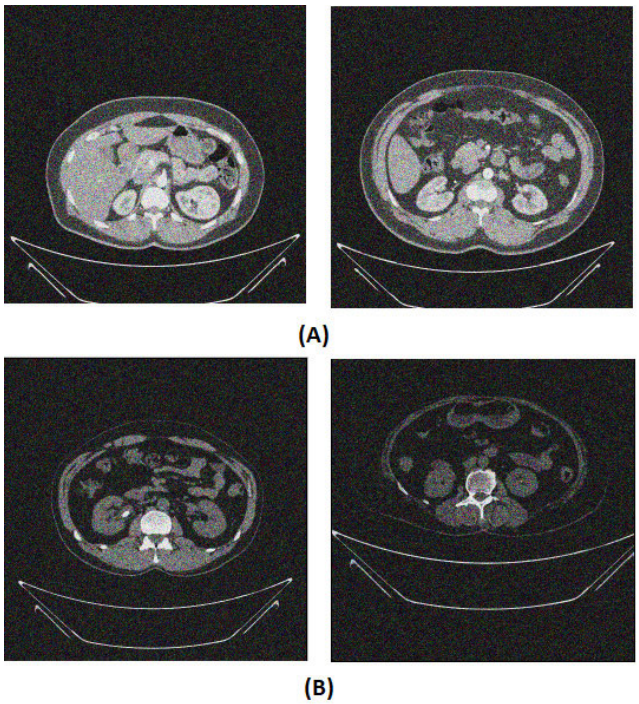


FIGURE 23. Some incorrectly classified test samples using the FINDWELL model from the KD3 dataset (A) Normal, (B) Kidney stone.

TABLE 9. Comparison of accuracy on KD dataset using different CV folds.

CV	Precision (%)
5-fold	66.2
10-fold	68.2
15-fold	67.6
20-fold	67.3
25-fold	66.8
30-fold	66.5

Before calculating all the performance measures using the KD dataset, we examined the precision values (as the dataset is imbalanced so accuracy is not a good choice) obtained from the proposed ensemble DNN using different CV folds i.e., {5, 10, 15, 20, 25 and 30} which is depicted in Table 9.

Table 9 shows that a 10-fold CV produces better precision results on the KD dataset compared to other folds.

Figure 24 depicts the confusion matrix produced using the FINDWELL model on the test samples of the KD dataset and the performance measure.

It can be noticed from the experimental results obtained from the performance of the proposed method on the KD dataset, that the performance is quite poor compared to the augmented datasets.

Fig. 25 depicts some correctly classified test samples using the FINDWELL model from the KD dataset.

Despite the poor performance of the FINDWELL model on the KD dataset, some images are correctly classified, as shown in Figure 25. The reason may be the proper use of K-fold cross-validation.

Fig. 26 depicts some incorrectly classified test samples using the FINDWELL model from the KD dataset.

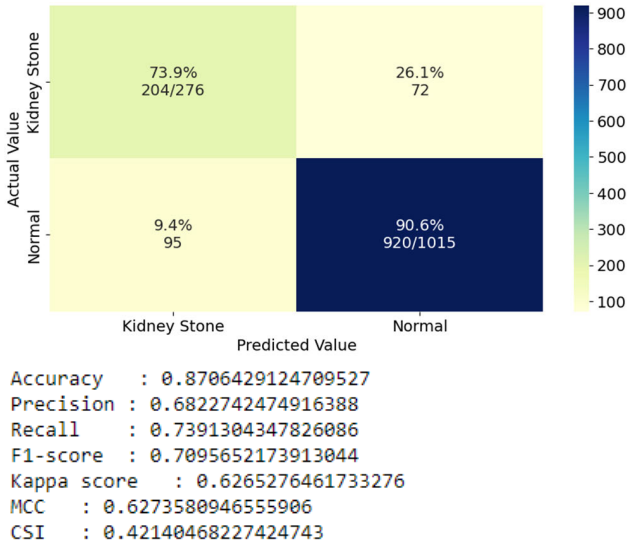


FIGURE 24. Confusion matrix and performance measure using the FINDWELL model when applied to the test samples of the KD dataset.

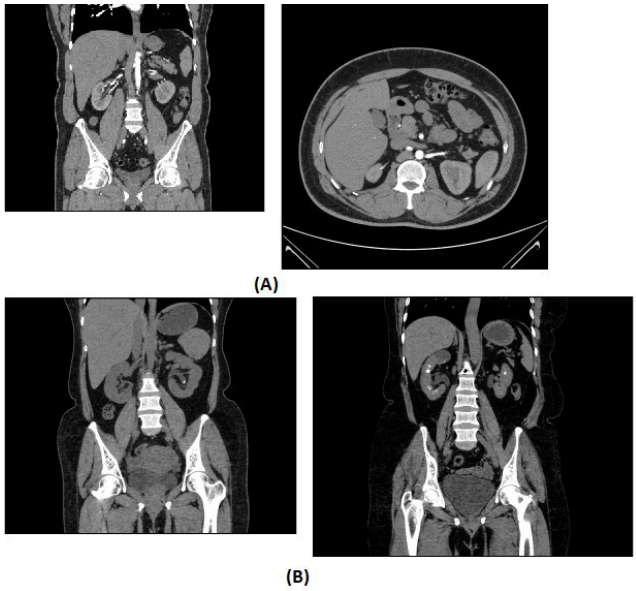


FIGURE 25. Some correctly classified test samples using the FINDWELL model from the KD dataset (A) Normal, (B) Kidney stone.

The main reason for the misclassification can be the proportion of normal and stone images which is imbalanced. This imbalance can cause problems in deep learning, as algorithms may fail to learn and predict accurately for the minority class, resulting in biased outputs and poor performance, which is visible.

K. PERFORMANCE COMPARISON OF THE PROPOSED FINDWELL MODEL WITH THE EXISTING METHODS

Table 10 compares the FINDWELL model's accuracy using a 10-fold CV with the existing method.

According to Table 10, the performance of the FINDWELL model is superior to the current methods for all three datasets.

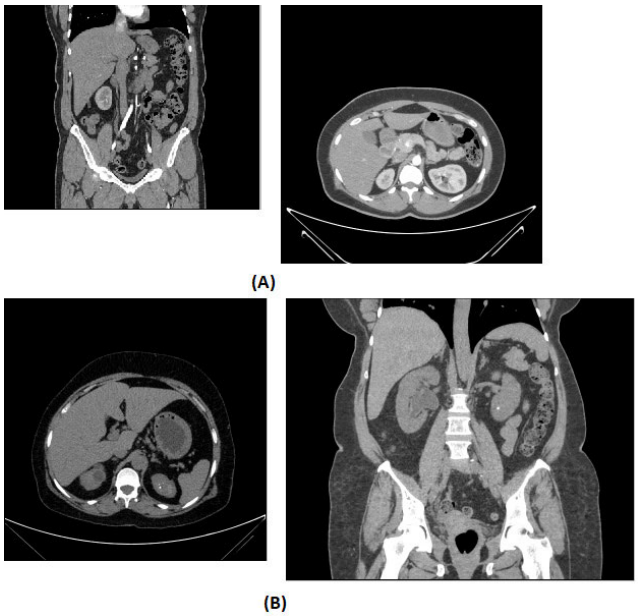


FIGURE 26. Some incorrectly classified test samples using the FINDWELL model from the KD dataset (A) Normal, (B) Kidney stone.

TABLE 10. The accuracy comparison (%) of the FINDWELL model with the existing method.

Dataset	Noise Level	AN	GN	DN169	MNV2	RN101
KD1	~	88.4	87.9	89.7	91.8	95.6
KD2	0.02	57.1	51.3	60.5	68.3	76.5
	0.04	51.6	46.1	56.7	59.2	70.7
	0.06	40.3	41.4	49.1	52.8	65.8
	0.08	35.4	34.7	41.6	45.1	59.2
	0.10	27.2	28.1	36.3	38.7	58.4
KD3	0.02	86.7	85.6	88.7	90.3	92.5
	0.04	84.3	83.4	85.1	90.1	90.3
	0.06	83.2	81.5	85.6	88.7	87.4
	0.08	78.1	77.3	79.7	84.5	86.7
	0.10	78.2	75.1	80.3	82.4	83.1

Dataset	Noise Level	SN	IV3	VGG16	ShN	DN19	PA
KD1	~	74.1	95.3	85.4	94.2	96.3	99.8
KD2	0.02	33.6	79.1	42.4	74.8	83.6	85
	0.04	27.4	72.6	37.3	62.1	77.4	76.6
	0.06	20.3	68.4	29.2	49.3	70.1	71.4
	0.08	12.1	61.3	23.7	41.4	67.5	64.7
	0.10	7.6	56.5	15.8	33.2	60.3	57.1
KD3	0.02	67.8	91.1	73.7	90.4	94.6	96.7
	0.04	61.6	90.3	72.1	88.3	91.5	96.2
	0.06	63.2	88.4	70.3	86.7	89.2	95.8
	0.08	56.3	87.7	65.8	85.5	87.8	95.1
	0.10	58.7	85.2	61.6	81.1	88.7	94.6

V. DISCUSSION

Disease locations in medical images are often visible in a limited portion of the image. Local traits must be taken from such localized locations to identify diseases properly. As a result, an ensemble deep feature creator is employed in the proposed model (FINDWELL) to make use of the benefits of more than one DNN. This effort also tries to use the benefits of inductive transfer. Pretrained DarkNet19, ResNet101, and InceptionV3 models make it simple to construct deep features. As a result, in this study, the ensemble feature vector is generated utilizing the models above. IRF feature selection is utilized to lower the time load of the utilized classifier while increasing the performance of CT kidney image detection.

TABLE 11. Performance comparison of the FINDWELL model on the KD, KD1, KD2, and KD3 dataset.

Data set	Performance Measure (in %)								
	A	P	R	F1	KS	MCC	GDR	CSI	PV
KD	87.0	68.2	73.9	70.9	62.6	62.7	39.5	42.1	1.0e-111
KD1	99.8	99.7	99.8	99.7	99.5	99.5	99.5	99.5	0.0
KD2	85.0	87.5	81.7	84.5	70.0	70.1	70	69.2	2.9e-258
KD3	96.7	97.8	95.5	96.6	93.3	93.3	93.3	93.3	0.0

*A: Accuracy, P: Precision, R: Recall, F1: F1 score, KS: Kappa Score, MCC: Matthews correlation coefficient, GDR: Good Detection Rate, CSI: Classification Success Index, PV: p-value

This study employs IRF, KNN, and BO approaches since they present a unique feature engineering model. The performance comparison of the FINDWELL model using the KD, KD1, KD2, and KD3 datasets is shown in Table 11.

From Table 11, we can deduce the following points regarding the performance efficiency of the FINDWELL model. We can conclude from the accuracy performance measure that the FINDWELL model can produce good classification results using quality and noisy images. From the precision, recall, and F1 score values obtained from the KD1 and KD3 datasets we can conclude that the model can efficiently detect the kidney stone CT images in both quality and noisy environments. After observing the kappa score, we can say that the FINDWELL model has a very good agreement between the real-world observer vs classification model. The high MCC value obtained denotes that the FINDWELL model has correctly classified a high percentage of normal CT images and a high percentage of kidney stone CT images. The high GDR value represents that the FINDWELL model is an efficient model detecting CT images in both quality and noisy environments. The high CSI value indicates that the FINDWELL model is well suited for detecting rare events in both quality and noisy environments. The p-value denotes that the proposed FINDWELL model can efficiently handle both KD1 and KD3 dataset images. The overall performance of the model using the KD dataset is very poor due to the imbalance characteristic of the dataset. The performance measures of the FINDWELL model using the KD2 dataset images are also poor compared to the KD1 and KD3 datasets. The reason can be the mismatch of the training and testing feature vectors (the training dataset doesn't contain any noisy samples, but we tested the performance of the FINDWELL model using noisy samples). So, we can say that the FINDWELL model will not perform well if the dataset is imbalanced or the testing samples do not have the same properties as the training samples.

Figures 13 and 21 show that the proposed FINDWELL model can efficiently classify both quality and noisy images.

Table 10 also shows comparative results achieved by the proposed model with previously utilized approaches. Table 10 clearly illustrates that the FINDWELL model outperformed other DNNs utilized in the literature regarding accuracy.

VI. CONCLUSION

We introduced a kidney stone detection approach in this paper that combined deep features from pre-trained deep convolutional neural networks with ML classifiers. To extract deep characteristics from kidney CT images, we suggest using many pre-trained deep convolutional neural networks. The retrieved deep features are subsequently analyzed using a KNN classifier that has been tuned by BO. The top three deep features that perform well on the KNN classifier are chosen and ensembled as a deep feature ensemble. Then IRF feature selector is used to use only the meaningful features for the classification which is then fed into the KNN classifiers to predict the final output. In this experiment, we performed a thorough study of kidney stone identification utilizing 10 different pre-trained deep convolutional neural networks on four distinct datasets (KD, KD1, KD2, and KD3).

Our experiment findings show that from our architecture (1) the DarkNet-19 deep feature alone is a good option in kidney stone detection, and (2) the ensemble of DarkNet19, ResNet101, and InceptionV3 deep features is a good choice in kidney stone detection, (3)Selection of relevant features using IRF can increase the model performance, (4) FINDWELL can detect both qualities and noisy images. In conclusion, our suggested FINDWELL ensemble technique overcomes the limits of a single CNN model and offers improved and robust performance, particularly for big datasets. These findings revealed that our proposed technique, which employs an ensemble of deep features and an ML classifier, might aid radiologists and nephrologists in identifying stones in CT kidney images. Modifications to the design of DNNs to increase classification accuracy appear to be an intriguing future effort.

The following are the limitations of the study. In this study, 6000 normal and 6000 patient individuals CT scans are used. The model must be evaluated with a more varied dataset to generalize the performance and get a robust model. We are unable to reveal the location of kidney stones in this work. An intelligent segmentation model can be used to solve this challenge. In the future, we hope to investigate the feasibility of employing the created FINDWELL model to diagnose various tumors and disorders using CT scans. The FINDWELL model may also be utilized as a learning model to tackle other computer vision challenges.

DECLARATION COMPETING INTERESTS

The authors declare no competing interests.

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